Name: _____

WORKSHEET

Converse

1 Write the converse of each statement.

a If an animal has wings, it can fly.

b If a shape has four sides, it is a quadrilateral.

c If you use the internet then you use Google as a search engine.

d If you go to high school then you went to primary school previously.

e If a graph is a parabola, it has a minimum.

f If $\frac{a}{b} < 1$, then $a < b$.

2 Given the true statements below, write the converse of each and decide whether or not the converse is also true.

a If it is snowing then it is cold.

b If something contains a lot of sugar it tastes sweet.

c If two triangles are congruent then their corresponding angles are equal in size.

d If a parabola passes through the origin, it has two x-intercepts.

e $x > y \Rightarrow y - x < 0$.

f If $0 < x < 1$, then $x^n < x$, where n $\in (1, \infty)$.

© Cengage Learning Australia Pty Ltd 2013

3 Decide whether equivalence holds for the statements below. If so, write $A \Leftrightarrow B$.

a A: an animal is a bird — B: it lays eggs

b A: February has 29 days — B: it is a leap year

c A: a triangle is equilateral — B: the triangle has three angles equal in size.

d A: a shape is a rhombus — B: its area can be calculated by multiplying the value of the base by the value of the height.

e A: the absolute value of an expression is required — B: the answer is positive

f A: $x = -2$ — B: $10 - 2x = 14$

4 For each true statement below, find the converse and decide whether or not it is an equivalence. If so, write it in the form P iff Q.

a If two angles are alternate then they are equal in size.

b If triangle DEF is scalene, its three angles are different sizes.

c If x is odd, then x^4 is also odd.

d If $a > 1$, then $\frac{1}{a} < 1$.

e If $x = -8$ then $x^2 + 5x = 24$.

f If p is divisible by 5, then p^2 is divisible by 25.

© Cengage Learning Australia Pty Ltd 2013

Answers

1 a If an animal can fly, it has wings.

b If a shape is a quadrilateral, then it has four sides.

c If you use Google as a search engine, then you use the internet.

d If you went to primary school previously, you now go to high school.

e If a graph has a minimum, it is a parabola.

f If $a < b$, then $\frac{a}{b} < 1$.

2 a If it is cold, then it is snowing. False.

b If something tastes sweet, it contains a lot of sugar. False.

c If two triangles have corresponding angles equal in size, they are congruent. False.

d If a parabola has two x-intercepts, it passes through the origin. False.

e $y - x < 0 \Rightarrow x > y$. $x > y$ iff $y - x < 0$.

f If $x^n < x$, where n $\in (1, \infty)$, then $0 < x < 1$. False (e.g. $x < 0$ and n is odd).

3 a $A \Rightarrow B$ is true. $B \Rightarrow A$ is not true as many other animals lay eggs. This is not an equivalence.

b $A \Leftrightarrow B$

c $A \Leftrightarrow B$

d $A \Rightarrow B$ is true. $B \Rightarrow A$ is not true as other shapes (e.g. rectangle) use this formula to calculate area. This is not an equivalence.

e $A \Rightarrow B$ is true. $B \Rightarrow A$ is not true many expressions can be positive without requiring the absolute value (e.g. $2x^2 + 4$, where $x \in R$). This is not an equivalence.

f $A \Leftrightarrow B$

4 a If two angles are equal in size, they are alternate. $P \Rightarrow Q$ is true. $Q \Rightarrow P$ is not true as other angle relationships (e.g. vertically opposite) have angles equal in size. This is not an equivalence.

b If triangle DEF has three angles different in size, it is scalene. P iff Q.

c If x^4 is odd, then x is also odd. P iff Q.

d If $\frac{1}{a} < 1$, then $a > 1$. $P \Rightarrow Q$ is true. $Q \Rightarrow P$ is not true as if a may be negative. This is not an equivalence.

e If $x^2 + 5x = 24$ then $x = -8$. $P \Rightarrow Q$ is true. $Q \Rightarrow P$ is not true as another solution exists ($x = 3$). This is not an equivalence.

f If p^2 is divisible by 25, then p is divisible by 5. P iff Q.

© Cengage Learning Australia Pty Ltd 2013